

XIAOFU HE, Ph.D.

**Assistant Professor of Clinical Neurobiology (In Psychiatry)
Columbia University
Research Scientist IV
Division of Child & Adolescent Psychiatry
New York State Psychiatric Institute**

Curriculum Vitae

Preparation Date of CV: September 20, 2025

Personal Data:

Xiaofu He
Office: 40 Haven Ave., Kolb 231B, New York, NY, 10032
Cell phone: 706-627-9973
Email: xh2170@cumc.columbia.edu
Profile: <https://datascience.columbia.edu/people/xiaofu-he/>
Website: <http://www.columbia.edu/~xh2170/>
Google Scholar: H-index 25
<https://scholar.google.com/citations?user=QJnxJFIAAAAJ&hl=en&oi=ao>

Academic Appointments

01/2018-present	Affiliated Faculty Member, Data Science Institute (DSI), Columbia University, NY
02/2018-present	Adjunct Assistant Professor, Department of Statistics, Columbia University, NY
06/2016-present	Research Scientist IV, Research Foundation for Mental Hygiene, Inc., the New York State Psychiatric Institute, NY
02/2014-present	Assistant Professor of Clinical Neurobiology (in Psychiatry), Department of Psychiatry, Columbia University, NY
02/2014-06/2016	Research Scientist III, Research Foundation for Mental Hygiene, Inc., New York State Psychiatric Institute, NY
06/2011-02/2014	Postdoctoral Research Scientist, Brain Imaging Lab, Department of Psychiatry, Columbia University, NY
02/2009-05/2011	Postdoctoral Research Fellow, Brain & Behavior Discovery Institute, Georgia Health Sciences University, GA
07/2007-02/2009	Assistant Professor, Department of Computer Science and Technology, East China Normal University, Shanghai, China

Education

09/2004-07/2007	Ph.D., Pattern Recognition and Intelligent Systems, Shanghai Jiao Tong University, Shanghai, China Thesis title: Research on Key Technology of Live Iris Recognition Advisor: Pengfei Shi
09/2001-06/2004	M.S., Computer Science, Shanghai Normal University, Shanghai, China

09/1998-06/2001 B.S., Computer Information Management, Fudan University, Shanghai, China

Postdoctoral Training

06/2011-02/2014 Postdoctoral Research Scientist, Brain Imaging Lab, Department of Psychiatry, Columbia University, NY
02/2009-05/2011 Postdoctoral Research Fellow, Brain & Behavior Discovery Institute, Georgia Health Sciences University, GA

Honors & Awards

2015 Irving Institute Imaging Pilot Award
2014 New York State Psychiatry Institute MRI Pilot Award
2013-2015 NARSAD Young Investigator Award
2010 Travel Award, SENN - GA/SC Neuroscience Consortium
2007 Best Paper Award (Second Author), Proceedings of the 6th National, Signal and Information Processing, Shanghai, China
2007 Outstanding Graduate of Shanghai, Shanghai, China
2006 STEP Electric Corporation Scholarship, Shanghai Jiao Tong University, Shanghai, China
2005 National Merit Scholarship, Shanghai Jiao Tong University, Shanghai, China
2006 Travel Award, Regional Inter-University Postgraduate Electrical and Electronic Engineering Conference (RIUPEEEEC 2005), Hong Kong, China

Administrative Leadership and Academic Service

2025-present Lead organizer, Inter-disciplinary Collaboration Club (ICC) Seminar Series, Columbia University, NY
2025-present Co-leader, the Neuroimaging Journal Club in the Department of Biostatistics in Psychiatry at Columbia University, NY
2024 Workshop co-chair, titled "Elucidation of mechanistic information using neuroimaging for psychiatric disorders", the 17th International Conference on Brain Informatics, Dec. 13-15, Bangkok, Thailand
2024 Technical Program Chair, 17th International Congress on Image and Signal Processing, Biomedical Engineering and Informatics (CISP-BMEI), Oct. 26-28, Shanghai China
2023 Program Committee, the 16th International Conference on Brain Informatics, Aug. 1-3, Hoboken, NJ
2019 Program Committee, New York Scientific Data Summit, June 12-14, NY
2018-present Committee member, Health Analytics, Data Science Institute, Columbia University, NY
2018-present Member, Computing Systems for Data-Driven Science, Data Science Institute, Columbia University, NY
2017-present Member, Foundations of Data Science, Data Science Institute, Columbia University, NY
2017 Program Committee for Real-time Functional Imaging and Neurofeedback (rtFIN), Nara, Japan

Professional Organizations and Societies

2013-present Senior member, Institute of Electrical and Electronics Engineers (IEEE)
2012-present Member, Organization for Human Brain Mapping (OHBM)

Editorial Board

2022-present Associate Editor in Meta-Radiology
2022-present Associate Editor in Frontiers in Psychiatry - Aging Psychiatry
2022 Guest editor in Post-Stroke Neural Plasticity: Functional and Structural Reorganization during Stroke Recovery 2022

Reviewer

European Journal of Pediatrics, Mega-Radiology, NeuroPlasticity, Neurolmage, Cerebral Cortex, PlosOne, Sensor Review, Pattern Analysis and Applications, Magnetic Resonance Imaging, International Journal of Biomedical Imaging, International Journal of Electronics, Frontiers, Applied Physics Letters, IEEE Transactions on Instrumentation and Measurement, IEEE Transactions on Biomedical Engineering, IEEE Transactions on Systems, Man and Cybernetics-Part C, IEEE Transactions on Automation Science and Engineering, The Organization for Human Brain Mapping

Invited lecture

July 17, 2025 – *International Symposium on Decoded Neurofeedback*, “Connectivity-Based and Decoded fMRI Neurofeedback in Youth and Adults with Mood and Psychotic Disorders”, Nara, Japan
April 24, 2025 – *Brain Imaging Lab (Miller Lab)*: “Real-time fMRI Neurofeedback – A Promising Treatment for Psychiatric Disorders”, Columbia University, NY
April 4, 2025 – *The Cognitive Development & Neuroimaging Laboratory (Marsh Lab)*: “Real-time fMRI Neurofeedback for Obsessive-Compulsive Disorder (OCD)”, Columbia University, NY
August 6, 2024 – *MRI unit, the New York State Psychiatric Institute*: “A Validation Test for Decoded fMRI Neurofeedback (DecNef)”, NY
June 27, 2024 – *Division of Child and Adolescent Psychiatry, Department of Psychiatry, Columbia University*: “Decoded fMRI Neurofeedback (DecNef) – A Promising Approach for Psychiatric Disorder Research”, NY
July 27, 2023 – *Xiang Li’s Lab, Massachusetts General Hospital*: “Machine Learning for Mental Health: Three Innovative Applications”, MA
November 4, 2022 – *Dr. Seonjoo Lee’s Lab, Department of Psychiatry, Columbia University*: “Introduction to Diffusion Weighted Imaging”, NY
July 16, 2021 – *GE HealthCare*: “The Challenges of Processing DTI Data Acquired with Diffusion Gradient Cycling”
February 14, 2020 – *Department of Psychology, Columbia University*: “Preprocessing of fMRI Data”, NY
February 7, 2020 – *Child Mind Institute*: “Applications of Machine Learning to Diffusion Tensor Imaging Data Analysis”, NY
January 13, 2020 – *Dr. Bret Rutherford’s Lab, Department of Psychiatry, Columbia University*: “Real-time fMRI Neurofeedback for Depression Research”, NY
November 25, 2019 – *Nathan Kline Institute (NKI)*: “Real-time fMRI Neurofeedback for Mental Disorders Research”, NJ
April 26, 2019 – *Inter-disciplinary Collaboration Club (ICC), Columbia University*: “Real-time Functional Neurofeedback for Mental Disorders”, NY
February 1, 2019 – *DSI Health Analytics Workshop, Columbia University*:

“Deep Learning for Mental Health: Three Innovative Applications”, NY

Fellowship and Grant Support

Active Research Funding

R01MH1234 87	Machine Learning Methods for Optimizing Individualized Treatment Strategies for Precision Psychiatry (PI: Yuanjia Wang)	Subcontract-Principal Investigator	2021-2026	NIH
Simon's Foundation	Targeted restoration of chromatin regulators for treatment of ASDs and developmental delay (PI: Bin Xu)	Collaborator	2021-2025	Simon's Foundation
U01-OH011694-01	Multimodal Neuroimaging of Cognitive and Emotional Networks in Young Adults Exposed to 9/11 as Children (MPI: Yael Cycowicz, Diana Moreno)	Co-Investigator	2018-2025	NIOSH-NCE
R01 MH114965	Deficient Belief Updating as a Convergent Computational Mechanism of Psychosis (PI: Guillermo Horga)	Collaborator	2018-2025	NIH-NCE

Past Support

K01 MH122774	Identification of Distinct Multimodal Biotypes of PTSD Using Data Driven Approach: A Multisite Big Data Study (PI: Xi Zhu)	Collaborator	2021-2024	NIH
R01MH1241 06	A data science framework for empirically evaluating addressing reproducible and transferable RDoC constructs in youth (MPI: Ying Liu, Seonjoo Lee)	Collaborator	2021-2024	NIH
R01 AG062578	Statistical method for neural mechanism mediating and moderating cognitive system in Alzheimer's disease and aging research (PI: Seonjoo Lee)	Collaborator	2020-2024	NIA

CUIT Emerging Technolo- gies Grant	Neuro-DBER: A flexible neuroscience- and technology-driven framework for Discipline-Based Education Research (MPI: Alfredo Spagna)	MPI	2023- 2024	Columbia University
1R33MH126 187-01	Optimizing placebo effects in depressed older adults: Enhancing processing speed and executive functioning with computerized cognitive training (PI: Bret Rutherford)	Co-Investigator	2021- 2023	NIH
R01 MH123660	Dopaminergic Dysfunction in Late-Life Depression (D3 Study) (PI: Bret Rutherford)	Co-Investigator	2020- 2023	NIH
R01 MH111596	Cognitive and Neural Mechanisms of the Accelerated Aging Phenotype in PTSD (PI: Bret Rutherford)	Collaborator	2017- 2023	NIH
Provost SOLER Seed Grants	Neuro-DBER: A New Area of Discipline-Based Education Research that Leverages Neuroscience to Enhance Teaching and Learning (MPI: Alfredo Spagna)	MPI	2022- 2023	Columbia University
Chairman's Collabo- rative Pilot Awards	Decoded fMRI neurofeedback for auditory hallucinations: a novel mechanistic and therapeutic tool targeting hallucination- related activation patterns (PI: Guillermo Horga)	Collaborator	2019- 2022	NYSPI/CU
R01- MH115024- 01A1	Task Control Circuit Targets for Obsessive Compulsive Behaviors in Children (PI: Rachel Marsh)	Technical Specialist	2018- 2022	NIH
Provost SOLER Seed Grants	Imaging the Brain Activity of Students Studying Brain Imaging (MPI: Alfredo Spagna)	MPI	2020- 2021	Columbia University

Collaborator Fellows Fund	Human Brain Imaging: Data, Analysis, and Sharing (MPI: Alfredo Spagna)	MPI	2019- 2021	Columbia University
R33 MH110029	Targeting Dopaminergic Mechanisms of Slowing to Improve Late Life Depression (PI: Bret Rutherford)	Collaborator	2016- 2021	NIH
CU	Irving Imaging Pilot Study (PI: Kimberly G. Noble)	Consultant	2016- 2019	CU
R01- DA038154- 01A1	Cognitive Control in Children of SUD Parents: A Longitudinal Multimodal MRI Study (PI: Christina Hoven)	Co-Investigator	2015- 2021	NIH
Young Investigator Grant	Common Structural and Functional Cortical Architecture in Children and Adolescents with Depression	Principal Investigator	2014- 2017	NARSAD
New York State Psychiatry Institute MRI Pilot Award	Connectivity-based real-time fMRI in youth with depression	Principal Investigator	2014- 2017	New York State Psychiatry Institute
Columbia Irving Institute Imaging Pilot Award	A Pilot Multimodal Imaging Study of Youth with Depression	Principal Investigator	2015- 2017	Columbia
R01 MH102293	Mechanisms of Antidepressant Non- Response in Late-Life Depression (PI: Bret Rutherford)	Collaborator	2014- 2019	NIH
R01- MH104648	Control and Reward Circuits as Targets for Repetitive Behaviors (PI: Rachel Marsh)	Collaborator	2014- 2018	NIH

R01-MH090062	A Longitudinal Multimodal MRI Study of Adolescents with Bulimia Nervosa (PI: Rachel Marsh)	Co-Investigator	2013-2018	NIH
R21-DA031975-01	SUD & Decision-Making: A Pilot Imaging Study of Parents & Children (PI: Christina Hoven)	Co-Investigator	2013-2016	NIH
R01-DA027100	Impact of Prenatal and Early Childhood Environmental Tobacco Smoke Exposure on Brain Development (PI: Bradley Peterson)	Postdoc Fellow	2011-2013	NIH
R01-MH089582	Identifying Brain-Based Biomarkers for ASD & their Biological Subtypes (PI: Bradley Peterson)	Postdoc Fellow	2011-2013	NIH
<u>Pending Funding</u>				
NIH R61/33 (scored 31)	Decoded fMRI neurofeedback for auditory verbal hallucinations in schizophrenia (MPI: Guillermo Horga, Joshua Kantrowitz)	MPI	2025-2030	NIH
SFARI	AI-based Sleep-Dependent EEG Biomarkers for Autism Spectrum Disorder	Principal Investigator	2026-2027	Simons
CU	AI-Based Biomarker Discovery from Multi-Night EEG for Sleep Disturbances in Autism Spectrum Disorder	Principal Investigator	2026-2027	CU
CU	The Contribution of Histone Modifications to the Epigenetic Landscape of Psychiatric Disorders (PI: Bin Xu)	Co-Investigator	2026-2027	CU
CU	Real-Time Glutamate Regulation as a Therapeutic Target in Translational Psychiatry (PI: Larry Kegeles)	Co-Investigator	2026-2027	CU

CU	Real-Time fMRI Neurofeedback Targeting Hippocampal–Cortical Connectivity to Enhance Memory Consolidation (PI: Lila Davachi)	Co-Investigator	2026- 2027	CU
CU	Neuromodulation of auditory verbal hallucinations in schizophrenia with decoded fMRI neurofeedback	Principal Investigator	2025- 2026	CU
CU	Neuromodulation of Emotion Regulation in Depression using Decoded fMRI Neurofeedback (MPI: Noam M. Schneck)	MPI	2025- 2026	CU

Pending Grant Submissions

R61/33	Connectivity-based Real-time fMRI Neurofeedback for OCD (MPI: Rachel Marsh, Kate Fitzgerald)	MPI	2026- 2031	NIH
Distinguishe d Investigator Grant	Real-Time Glutamate Neurofeedback Regulation as a Therapeutic Target (PI: Larry Kegeles)	Co-Investigator	2026- 2027	BBRF
R21	Decoded fMRI neurofeedback for Emotion Regulation in Depression (MPI: Noam M. Schneck)	MPI	2026- 2028	NIH
R21	Connectivity-Based Real-time fMRI Neurofeedback for Enhancing Hippocampal- Cortical Connectivity (MPI: Lila Davachi)	MPI	2026- 2028	NIH
R21	Developing a Generalized Deep Learning Framework for Diffusion Tensor Imaging Preprocessing	Principal Investigator	2026- 2028	NIH

Educational Contributions

Direct Teaching

Fall	2025	<i>Applied Machine Learning for Computer Vision</i> , MS students, Data Science Institute & Department of Statistics, Columbia University
Fall	2024	<i>Applied Machine Learning for Computer Vision</i> , MS students, Data Science Institute & Department of Statistics, Columbia University
Fall	2023	<i>Applied Machine Learning for Computer Vision</i> , MS students, Data Science Institute & Department of Statistics, Columbia University
Fall	2022	<i>Applied Machine Learning for Image Analysis</i> , MS students, Data Science Institute & Department of Statistics, Columbia University
Fall	2021	<i>Applied Machine Learning for Image Analysis</i> , MS students, Data Science Institute & Department of Statistics, Columbia University
Fall	2021	<i>Fundamentals of human brain imaging: from theory to practice</i> , Undergraduate students, Department of Psychology, Columbia University (co-teaching with Dr. Alfredo Spagna)
Fall	2020	<i>Applied Machine Learning for Image Analysis</i> , MS students, Data Science Institute & Department of Statistics, Columbia University
Fall	2019	<i>Applied Machine Learning for Image Analysis</i> , MS students, Data Science Institute & Department of Statistics, Columbia University
Fall	2018	<i>Applied Machine Learning for Image Analysis</i> , MS students, Data Science Institute & Department of Statistics, Columbia University
Spring	2018	<i>Applied Machine Learning for Image Analysis</i> , MS students, Data Science Institute & Department of Statistics, Columbia University
Spring	2018	<i>Introduction to Brain Computer Interaction</i> , Undergraduate and MS students, Department of Computer Science, Columbia University
2008-2009		<i>Robot Technique and Application</i> , Undergraduate students, Department of Computer Science, School of Continuing Education, Shanghai Jiao Tong University
2007-2009		<i>Image Processing and Computer Vision</i> , Undergraduate students, Department of Computer Science and Technology, East China Normal University
2003-2004		<i>Visual C++ Development and Programming in C</i> , Undergraduate students, Department of Aptech Certified Computer Professional, Shanghai Science & Technology College

Advising and Mentorship

Ph.D. Trainees

2022-2024	Josh Sterling Friedman, Project: Neuro-DBER: A flexible neuroscience- and technology-driven framework for Discipline-Based Education Research (joined our CU pilot study, co-PI: Alfredo Spagna, Ph.D., my role: EEG data acquisition, processing, and analysis). One first-authored poster, one co-authored poster, and one first-authored journal paper (under review).
2020-2023	Katrina Simon, Project: <i>Socioeconomic Factors, Stress, Hair Cortisol, and White Matter Microstructure in Children</i> (mentor: Kimberly G. Noble, Ph.D., my role: MRI data processing and analysis training). Three first-authored and three co-authored journal papers. Two first-authored abstracts.

2021-2022	Qinxia Wang, A latent state space model for estimating brain dynamics from electroencephalogram (EEG) data. (mentor: Yuanjia Wang, Ph.D., my role: EEG data processing and analysis training). One first-authored journal paper.
2018-2020	Qian Wang, visiting Ph.D. student. Project: Using machine learning methods to diagnose autism based on Diffusion tensor imaging (DTI) and functional magnetic resonance imaging (fMRI), one first-authored poster.
2017-2018	Zhenghua Hou, visiting Ph.D. student. Project: The predictive value of neuroimaging features on identifying early antidepressant response. Two first-authored and one co-authored journal papers. He is now an Assistant Professor at Southeast University.

Other Trainee Advisory Roles

Research Supervision of Visiting Faculty

2016-2017	Zhi Liu, Ph.D., Associate Professor, Project: <i>Machine Learning in Image Analysis</i> . NYSPI. One first-authored journal paper & one co-authored journal paper, along with two co-authored abstracts.
-----------	--

Technical Supervision of Post-Doctoral Research Fellows

2024-present	Josh Sterling Friedman, Project: Developing scalable frameworks for predicting student attention in the classroom with machine learning models and multimodal data from the body, heart, and brain (mentor: Alfredo Spagna, Ph.D., Columbia University). One first-authored paper (under review).
2023-present	Caitlin Lloyd, Ph.D., Project: Diffusion MRI preprocessing (mentor: Joanna E. Steinglass, MD, NYSPI). One co-authored an OHBM conference poster.
2016-2019	Emily C. Merz, Ph.D., Project: <i>Anxiety, depression, impulsivity, and brain structure in children and adolescents</i> (mentor: Kimberly G. Noble, Ph.D., Columbia University). Five first-authored and four co-authored journal papers, along with three first-authored and five co-authored abstracts. She is now an Assistant Professor at Colorado State University.
2015-2017	Mirjana Domakonda, M.D., Project: <i>functional Connectivity in Adolescents with Bulimia Nervosa</i> (mentor: Rachel Marsh, Ph.D., NYSPI). One first-authored journal paper and one oral presentation. She is now a psychiatrist at Hartford Hospital.
2015-2015	Lisa Pao, Ph.D., Project: <i>resting state fMRI data processing and analysis</i> (mentor: Rachel Marsh, Ph.D., NYSPI).

Technical Supervision of Research Assistants

2023-present	Luyu Huang, Collaborated with Dr. Xi Zhu on a PTSD project. NYSPI.
2022-present	Nicholas Bustos, Collaborated with Dr. Rachel Marsh on an OCD project. NYSPI.
2019-2022	Trained all research assistants for my collaborator, Dr. Bret Rutherford's lab, including monthly MRI data acquisition and quality control. NYSPI.

2021-2022	Caleb R. Haynes, MRI data organization, task-fMRI data preprocessing and analysis. Collaborated with Dr. Bret Rutherford's lab, NYSPI.
2017-2017	Michael Wang, fMRI data preprocessing and analysis. Collaborated with Dr. Christina Hoven at NYSPI.
2014-2019	Mihaela Stefan, quality control of DTI data preprocessing and analysis. Collaborated with Dr. Rachel Marsh's lab at NYSPI.
2014-2019	Martine Fontaine, quality control of DTI data preprocessing and analysis. Collaborated with Dr. Rachel Marsh's lab at NYSPI.

Research Supervision of Graduate Students

2025-present	Yunhao Luo, Department of Statistics, Columbia University. Project: Seizure Prediction from Mouse EEG Data.
2025-present	Junze Huang, Department of Statistics, Columbia University. Project: Decoded fMRI neurofeedback.
2025-present	Zanxu Wang, Department of Statistics, Columbia University. Project: Calcium image data processing and analysis.
2025-present	Yanhao Feng, Department of Statistics, Columbia University. Project: Real-time face and pose estimation.
2025-present	Ruoxuan Li, Data Science Institute, Columbia University. Project: Real-time EEG neurofeedback. One co-authored poster
2024-2025	Yutong Gao, Department of Statistics, Columbia University. Project: Calcium image data processing and analysis. One co-authored short paper.
2024-2024	Yixuan Wang, Department of Statistics, Columbia University. Project: White Matter Integrity in Youth with Suicide. Ideation/Suicide Attempt, Major Depressive Disorder, and Comorbidities. One first-authored conference paper. She is preparing for her PhD program application.
2023-present	Qifan Jiang, Data Science Institute, Columbia University. Project: Automatic Calcium imaging data preprocessing and analysis pipeline development.
2023-present	Hengbo Tong, Department of Statistics, Columbia University. Project: Neuro-DBER: A flexible neuroscience- and technology-driven framework for Discipline-Based Education Research
2023-2024	Helin Wang, Department of Psychology, NYU. Project: Characterizing the relationship between emotional, cognitive, and behavioral indices of students engagement in the classroom to pave the way for real-time intervention on self-regulation strategies.
2023-2024	Robert AJ Barry, Department of Statistics, Columbia University. Project: Deep learning based prediction of Autism using resting-state fMRI data. One co-authored arxiv paper. He is now a Founder, at Stealth Startup. San Francisco, CA.
2023-2024	Jingru Gong, Department of Computer Science, CMU. Project: real-time EEG/fMRI neurofeedback. Now she is a senior data analyst at Kohl.
2023-2023	Megan Zhou, Data Science Institute. Project: Decoded EEG neurofeedback. She is now a Data Scientist at Verisk Analytics, NJ.

2022-2023	Kaiyu He, Department of Biostatistics. Project: Deep Learning based EEG to fMRI Transformation. One co-authored poster.
2022-2023	Lihui Pan, Department of Statistics. Project: Diffusion Tensor Imaging Quality Assessment using CNN.
2022-2023	Siqi He, Data Science Institute. Project: image registration for diffusion tensor imaging. One co-authored poster. She is now a Project Manager at the Data Center of BoCOM, Shanghai, China.
2021-2023	Xuchen Wang, Data Science Institute. Project: Automation of Calcium Imaging Data Analysis pipeline.
2021-2024	Xinyuan Zheng, Department of Statistics. Project: ResNet for Autism classification. One co-authored journal paper and one first-author arxiv paper, and one first-authored poster. She is now a Ph.D. student in the Department of BME at Yale University, CT.
2021-2022	Junsheng Shi, Data Science Institute. Project: real-time EEG neurofeedback using Muse2. One first-authored poster and two co-authored posters. He is now a data scientist at Lazard Inc., NY.
2021-2022	Ming Li, Department of Statistics. Project: Deep Learning based EEG to fMRI Transformation. One first-authored poster. He is now a Founder at Forest Vision LLC, CA.
2021-2022	Xiaodi Sun, Department of Statistics. Project: Deep Learning based EEG to fMRI Transformation. One second-authored poster. He is now a Principal Applied Scientist Manager at Microsoft AI, WA.
2021-2022	Zhuowei Gu, Department of Computer Science and Engineering at New York University. Project: EEG data preprocessing using Muse 2. Two co-authored posters. He is now a PhD student in the Department of Computer Science at Saint Louis University, MO.
2021-2022	Hui Xiang, Department of Statistics. Project: EEG data preprocessing using Muse2. One co-authored poster. She is now a Standardized Trust Products Manager at AJ Trust Co, Ltd., Shanghai, China.
2020-2023	Yoojean Kim, Department of Biostatistics. Project: Predicting treatment response of Patients with Major Depressive Disorders based on neuroimaging data. Four co-authored journal papers and one co-authored arXiv paper. One first-authored poster and one co-authored poster. She is now a PhD student at CUNY Graduate Center, NY.
2020-2022	Zijing Wang, Department of Biostatistics. Project: Diffusion Tensor Imaging Quality Assessment. One co-authored arXiv paper and one co-authored poster. She is now a Data Scientist at MongoDB, NY.
2019-2021	Maxime Tchibozo, Data Science Institute. Project: Emotional <i>Brain State Classification using Convolutional Neural Networks</i> . One first-authored poster and one first-authored arXiv paper. He is now a financial analyst at Lazard in London, UK.
2019-2020	Xiaoyue Zhang, Department of Biostatistics. Project: Predicting subjects with a history of Major Depressive Disorder based on resting state fMRI data. One first authored poster. She is now a Data Scientist at Google, MA.

2019-2020	Donggeun Kim, Department of Biostatistics, Project: <i>Emotional Brain State Classification using Hyperalignment</i> . One co-authored poster and one co-authored arXiv paper. He is now a PhD candidate in Statistics at the University of Oxford, UK.
2019-2019	Xiaoqi Wang, Data Science Institute, Project: <i>Emotional Pictures Reconstruction based on BOLD signal using Deep Adversarial Neural Decoding</i> . She is now a data analyst at Facebook, NY.
2018-2021	Yijing Feng, Department of Electrical Engineering. Project: <i>Develop a platform for Real-time Brain State Classification on Mobile EEG</i> . One first-authored abstract. She is now a Ph.D. student in the Department of Computer Science at Dartmouth, NH.
2018-2019	Euiyoung Chung, Department of Statistics. Project: <i>Brain State Classification on EEG data</i> . One second-authored poster and one second-authored abstract. He is now a quantitative analyst at Millennium Management LLC, NY.
2018-2019	Peilin Li, Department of Statistics. Project: <i>Quality Control of Diffusion Tensor Imaging</i> . He is now a data analyst at Alibaba, China.
2018-2019	YoungCho (Andrew) Lee, Department of Statistics. Project: <i>Real-time Brain State Classification on Functional Near Infrared Spectroscopy</i> . One first-authored poster. He is now a software engineer at Amazon, TX.
2018-2019	Jongwoo Choi, Department of Statistics. Project: <i>Imaging-Psychiatry Challenge (IMPAC): predicting autism: A data challenge on Autism Spectrum Disorder detection</i> . One first-authored abstract. He is now a PhD student in the Department of Statistics at the University of Connecticut, CT.
2018-2019	Xizhu Deng, Department of Statistics. Project: <i>Real-time Brain State Classification on fMRI</i> . He is now a software engineer at Uber, TX.
2018-2018	Xinhui Li, Department of Biomedical Engineering. Project: <i>Visualization of Real-Time Functional Neurofeedback</i> . One co-authored abstract. She is now a PhD student in the Department of Computer Science at Georgia Institute of Technology, GA.
2017-2017	Hongyue Lan, Data Science Institute. Project: <i>Develop a Voice Assistant App for Android Smart Phones</i> . One demo App developed.
2016-2018	Sikai Huang, Department of Computer Science. Project: <i>Deep Learning in Brain Imaging</i> . One second-authored poster. He is now a Ph.D. student at the National University of Singapore, Singapore.
2016-2016	Kainan Zhang, MS, Research Assistant. Project: <i>Quality Control for Diffusion Tensor Imaging</i> . He is now a PhD student in the Department of Computer Science at Georgia State University, GA.

Research Supervision of Undergraduate Students

2023-2024	Rulin Ma, Department of Computer Science, NYU, Project: <i>Image Registration for Diffusion Tensor Imaging</i> . She is now an AI scientist at CGI Inc., Canada.
-----------	--

2022-2024	Conor Lee Shatto, Department of Psychology, Columbia University, Project: Neuro-DBER: A flexible neuroscience- and technology-driven framework for Discipline-Based Education Research. He is now a Project Manager at the PAIR Center, UPenn, PA.
2022-2025	Yian Wang, Department of Computer Science, UIUC. Project: <i>Image Registration for Diffusion Tensor Imaging</i> . He is now a MA student in the Department of Computer Science at UCSD, CA.
2020-2021	Isabella E. Rosario, Project: <i>Imaging the Brain Activity of Students Studying Brain Imaging</i> , Department of Psychology, Columbia University. She is now a PhD student in the Department of Neuroscience at Columbia University, NY.
2019-2020	Zhuowei Gu, Project: <i>Building a Docker Container for Diffusion Tensor Imaging Data Processing and Analysis Tools</i> , Department of Computer Science at Miami University at Oxford, Ohio. He is now an MS student in Computer Science at NYU, NY.
2020-2020	Lingbin Sun, Project: <i>Chatbot App with an Application on Mental Health</i> , Department of Computer Science at Purdue University, Indiana. She is now a software engineer, China.
2018-2018	Louis Kernel, Project: <i>Diffusion Tensor Imaging Data Processing in Python</i> , Department of Computer Science at The State University of New York at Buffalo, NY.
2007-2009	Advised eight undergraduate students' theses in computer vision and image processing, Department of Computer Science, East China Normal University, Shanghai, China.

Publications

Original, Peer Reviewed Articles

1. **He X**, Wang Y, Gao Y, Wang X, Sun Z, Zhu H, Leong K, Xu B (2025), CalciumZero: a toolbox for fluorescence calcium imaging on iPSC derived brain organoids, *Brain Informatics*, (2025) 12:2. <https://doi.org/10.1186/s40708-024-00248-5>.
2. Cheslack-Postava K, Cykowicz YM, Rodriguez-Moreno DV, Amsel LV, Musa GJ, Ryan M, Bresnahan M, Tang H, Geronazzo-Alman L, Bisaga A, Wang Z, **He X**, Hoven CW (2024), Characteristics of Adolescents with and without a Family History of Substance Use Disorder from a Minority Cohort. *Children (Basel)*. 2024 May 31;11(6):671. doi: 10.3390/children11060671.
3. **Kim YG, Ravid O, Zheng X, Kim Y, Neria Y, Lee S, He X*, Zhu X* (2024), Explaining deep learning-based representations of resting state functional connectivity data: focusing on interpreting nonlinear patterns in autism spectrum disorder. *Frontiers in Psychiatry*. 2024 May 20;15:1397093. doi: 10.3389/fpsy.2024.1397093. (*corresponding author)**
This study investigated latent contribution scores at the network level in individuals with autism spectrum disorder (ASD) and healthy controls. While both groups shared key network contributors—such as the ventral attention network (VAN) driving latent component 0—they exhibited distinct differences, including divergent VAN (latent 0) and sensory/somatomotor network (latent 2) scores. By introducing an interpretable variational autoencoder (VAE) framework, the study identified both shared and ASD-specific patterns of functional connectivity. These findings offer deeper insights into the neural underpinnings of autism. The integration of explainable AI tools with multimodal VAE models enhances model transparency

and diagnostic accuracy, paving the way for interpretable, AI-driven clinical tools and underscoring the transformative potential of deep learning in personalized mental health diagnostics and treatment.

I am the corresponding author and served as a collaborator on the K01 project, "Identification of Distinct Multimodal Biotypes of PTSD Using Data Driven Approach: A Multisite Big Data Study" (PI: Xi Zhu), which provided funding for this paper. I co-designed the study with Dr. Xi Zhu and contributed to the writing and editing of the manuscript.

4. Hansen M, Simon KR, **He X**, Steele N, Thomas ML, Noble KG, Merz EC (2024), Socioeconomic factors, sleep timing and duration, and amygdala resting-state functional connectivity in children. *Frontiers in Psychiatry*. 2024 May 22;15:1373546. doi: 10.3389/fpsy.2024.1373546. eCollection 2024.
5. Huggins AA, Baird CL, Briggs M, Laskowitz S, Hussain A, Fouda S, Haswell C, Sun D, Salminen LE, Jahanshad N, Thomopoulos SI, Veltman DJ, Frijling JL, Olff M, van Zuiden M, Koch SBJ, Nawijn L, Wang L, Zhu Y, Li G, Stein DJ, Ipser J, Seedat S, du Plessis S, van den Heuvel LL, Suarez-Jimenez B, Zhu X, Kim Y, **He X**, Zilcha-Mano S, Lazarov A, Neria Y, Stevens JS, Ressler KJ, Jovanovic T, van Rooij SJH, Fani N, Hudson AR, Mueller SC, Sierk A, Manthey A, Walter H, Daniels JK, Schmahl C, Herzog JI, Říha P, Rektor I, Lebois LAM, Kaufman ML, Olson EA, Baker JT, Rosso IM, King AP, Liberzon I, Angstadt M, Davenport ND, Sponheim SR, Disner SG, Straube T, Hofmann D, Qi R, Lu GM, Baugh LA, Forster GL, Simons RM, Simons JS, Magnotta VA, Fercho KA, Maron-Katz A, Etkin A, Cotton AS, O'Leary EN, Xie H, Wang X, Quidé Y, El-Hage W, Lissek S, Berg H, Bruce S, Cisler J, Ross M, Heringa RJ, Grupe DW, Nitschke JB, Davidson RJ, Larson CL, deRoos-Cassini TA, Tomas CW, Fitzgerald JM, Blackford JU, Olatunji BO, Kremen WS, Lyons MJ, Franz CE, Gordon EM, May G, Nelson SM, Abdallah CG, Levy I, Harpaz-Rotem I, Krystal JH, Dennis EL, Tate DF, Cifu DX, Walker WC, Wilde EA, Harding IH, Kerestes R, Thompson PM, Morey R (2024), Smaller total and subregional cerebellar volumes in posttraumatic stress disorder: a mega-analysis by the ENIGMA-PGC PTSD workgroup, *Molecular Psychiatry*. 2024 Mar;29(3):611-623. doi: 10.1038/s41380-023-02352-0.
6. Zhu X; Kim Y; Ravid O; **He X**, Benjamin SZ; Sigal ZM; Lazarov A; Lee S; Abdallah CG; Angstadt M; Averill CL; Baird CL; Baugh LA; Blackford JU; Bomyea J; Bruce SE; Bryant RA; Cao Z; Choi K; Cisler J; Cotton AS; Daniels JK; Davenport NK; Davidson RJ; DeBellis MD; Dennis EL; Densmore M; deRoos-Cassini T; Disner G; Hage WE; Etkin A; Negar Fani N; Fercho KA; Fitzgerald J; Forster GL; Frijling JL; Geuze E; Gonenc A; Gordon EM; Gruber S; Grupe DW; Guenette JP; Haswell CC; Heringa RJ; Herzog J; Hofmann DB; Hosseini B; Hudson AR; Huggins AA; Ipser JC; Jahanshad N; Richards MJ; Jovanovic T; Kaufman ML; Kennis M; King A; Kinzel P; Koch SBJ; Koerte IK; Koopowitz SM; Korgaonkar MS; Krystal JH; Lanius R; Larson CL; Lebois LAM; Li G; Liberzon I; Lu GM; Luo Y; Magnotta VA; Manthey A; Maron-Katz A; May G; McLaughlin K; Mueller SC; Nawijn L; Nelson SM; Neufeld RWJ; Nitschke JB; O'Leary EM; Olatunji BO; Olff M; Peverill M; K Phan L; Qi R; Quidé Y; Rektor I; Ressler K; Riha P; Ross M; Rosso IM; Salminen LE; Sambrook K; Schmahl C; Shenton ME; Sheridan M; Shih C; Sicorello M; Sierk A; Simmons AN; Simons RM; Simons JS; Sponheim SR; Stein B; Stein DJ; Stevens JS; Straube T; Sun D; Théberge J; Thompson PM; Thomopoulos SI; Wee NJA; Werff SJA; Erp TGM; Rooij SJH; Zuiden MV; Varkevisser T; Veltman DJ; Vermeiren

- RRJM; Walter H; Wang L; Wang X; Weis C; Winternitz S; Xie H; Zhu Y; Wall M; Neria Y; Morey RA (2023), Neuroimaging-Based Classification of PTSD Using Data-Driven Computational Approaches: A Multisite Big Data Study from the ENIGMA-PGC PTSD Consortium, *NeuroImage*. doi: 10.1016/j.neuroimage.2023.120412. Epub 2023 Oct 18.
7. Hansen M, Simon K, Strack J, **He X**, Noble K, Merz E (2023), Socioeconomic Disparities in Sleep Duration are Associated with Cortical Thickness in Children, *Brain and Behavior*. 2023 Feb;13(2): e2859. doi: 10.1002/brb3.2859. Epub 2022 Dec 27.
8. Wang Q, Loh JM, **He X**, Wang Y (2022), A latent state space model for estimating brain dynamics from electroencephalogram (EEG) data. *Biometrics*. <https://doi.org/10.1111/biom.13742>
9. Wang Z, Rodriguez Moreno DV, Cycowicz YM, Amsel LV, Cheslack-Postava K, **He X**, Ryan M, Geronazzo-Alman L, Musa GJ, Bisaga A, Hoven CW (2022). Shapes of Subcortical Structures in Adolescents with and Without Familial History of Substance Use Disorder. *Human Brain Mapping*, 43(9): 2759-2770. doi: 10.1002/hbm.25804.
10. Hou Z, Li T, **He X**, Li T, Zhang Y, Chen H. Jiang W, Y Yin, Yuan Y (2022), Distinct Features of Cerebral Blood Flow and Spontaneous Neural Activity as Integrated Predictors of Early Response to Antidepressants. *Frontiers in Psychiatry*, 18 January 2022 <https://doi.org/10.3389/fpsyt.2021.788398>.
11. Clausen AN, Fercho KA, Monsour M, Disner S, Salminen L, Haswell CC, Rubright EC, Watts AA, Buckley MN, Maron-Katz A, Sierk A, Manthey A, Suarez-Jimenez B, Olatunji BO, Averill CL, Hofmann D, Veltman DJ, Olson EA, Li G, Forster GL, Walter H, Fitzgerald J, Théberge J, Simons JS, Bomyea JA, Frijling JL, Krystal JH, Baker JT, Phan KL, Ressler K, Han LKM, Nawijn L, Lebois LAM, Schmaal L, Densmore M, Shenton ME, van Zuiden M, Stein M, Fani N, Simons RM, Neufeld RWJ, Lanius R, van Rooij S, Koch SBJ, Bonomo S, Jovanovic T, deRoos-Cassini T, Ely TD, Magnotta VA, **He X**, Abdallah CG, Etkin A, Schmahl C, Larson C, Rosso IM, Blackford JU, Stevens JS, Daniels JK, Herzog J, Kaufman ML, Olf M, Davidson RJ, Sponheim SR, Mueller SC, Straube T, Zhu X, Neria Y, Baugh LA, Cole JH, Thompson PM, Morey RA. Assessment of brain age in posttraumatic stress disorder: Findings from the ENIGMA PTSD and brain age working groups. *Brain Behavior*. 2022 Jan;12(1):e2413. doi: 10.1002/brb3.2413. Epub 2021 Dec 14. PubMed PMID: 34907666; PubMed Central PMCID: PMC8785613.
12. Simon K, Merz E, **He X**, Desai P, Meyer J, Maskus E, Noble K (2021), Socioeconomic Factors, Stress, Hair Cortisol, and White Matter Microstructure in Children. *Developmental Psychobiology*. PMID: 34105766 DOI: 10.1002/dev.22147
13. **He X, Morenob DR, Cycowicz Yael, Wang Z, Tang H, Amsel L, Cheslack-Postavaa K , Ryan M, Geronazzo-Alman L, Musa G, Bisaga A, Hoven CW (2021). White Matter Integrity and Functional Connectivity in Adolescents with a Parental History of Substance Use Disorder. *Neuroimage: Reports*. <https://doi.org/10.1016/j.ynirp.2021.100037>**
This study examined brain structure and function in adolescents with a family history (FH+) and without (FH-) a parental substance use disorder (SUD). While no differences were found in white matter integrity between groups, FH+ adolescents showed significantly greater functional connectivity between the posterior Default Mode Network (DMN) and the Fronto-Parietal Network (FPN). Although both groups reported similar levels of reward-seeking behavior, FH+ youth exhibited higher impulsivity, as measured by the BIS-11. However, impulsivity scores were

not significantly associated with either brain connectivity or white matter metrics. These findings suggest that increased DMN–FPN connectivity may represent a neural marker of inherited SUD risk and highlight the value of early intervention to support healthy brain development before substance use begins.

I was a co-Investigator (co-I) on the NIH R01 “Cognitive Control in Children of SUD Parents: A Longitudinal Multimodal MRI Study” (PI: Christina Hoven) that generated the data used in this paper. I contributed to the study's conception and planning as a co-I, data preprocessing and analyses on both DTI and resting-state fMRI data, and the writing and editing of the manuscript.

14. Ramphal B, Pagliaccio D, Thomas V. L, **He X**, Margolis E. A (2021), Contributions of Cerebellar White Matter Microstructure to Social Difficulty in Nonverbal Learning Disability. *The Cerebellum*. <https://doi.org/10.1007/s12311-021-01265-4>.
15. **He X, Stefan M, Pagliaccio D, Khamash L, Fontaine M, Marsh R (2021). A Quality Control Pipeline for Probabilistic Reconstruction of White-matter Pathways. *Journal of Neuroscience Methods*, 353 (2021): 109099, PMID: 33582173. <https://doi.org/10.1016/j.jneumeth.2021.109099>**
This paper presents a semi-automated quality control (QC) pipeline for TRACULA, a widely used tool in DTI analysis for reconstructing white matter tracts. While TRACULA reliably reconstructs 18 major white matter tracts, partial or incomplete reconstructions can occur, often leading to biased analyses or outright exclusion of data. The proposed pipeline integrates both automated and minimal manual steps: visual inspection of eddy-current corrected and color-encoded FA images, automated tract-volume checks, reprocessing of tracts below a volume threshold, manual control-point editing when needed, followed by TRACULA reinitialization. This approach enables users to efficiently evaluate the quality of reconstructed white matter tracts and provides a semi-automated solution for restoring incomplete tractography results.

I was a Technical Specialist on the NIH R01 “Task Control Circuit Targets for Obsessive Compulsive Behaviors in Children” (PI: Rachel Marsh) that generated data for this paper. I conceptualized, designed, and developed the QC pipeline, and contributed to the writing and editing of the manuscript.

16. **He X., Pueraro E., Kim Y., Garcia C.M, Maas B, Choi J., Egglefield D.A., Schiff S., Sneed J.R., Brown P.J., Brickman A.M., Roose S.P., Rutherford B.R. (2021). Association of White Matter Integrity with Executive Function and Antidepressant Treatment Outcome in Patients with Late-Life Depression, *The American Journal of Geriatric Psychiatry*, doi: 10.1016/j.jagp.2021.01.004**
This study investigated how white matter integrity, measured by fractional anisotropy (FA) using diffusion tensor imaging (DTI), relates to cognitive function and antidepressant treatment outcomes in older adults with late-life depression (LLD). Seventy-one participants underwent baseline neuroimaging and cognitive testing as part of an 8-week randomized controlled trial. Higher FA in the bilateral superior longitudinal fasciculus (SLF), SLF-temporal tracts, and the right corticospinal tract (CST) was associated with better performance on executive function tasks, specifically the Initiation/Perseveration subscale of the Dementia Rating Scale. Exploratory analyses further showed that increased FA in the right CST and right SLF regions predicted greater symptom improvement and treatment

response. These findings highlight the potential of DTI-based markers in identifying neurobiological predictors of cognitive function and treatment success in LLD.

I was a collaborator on the NIH R01 “Targeting Dopaminergic Mechanisms of Slowing to Improve Late Life Depression” (PI: Bret Rutherford), which provided the data for this paper. I contributed to the DTI data preprocessing and analysis, statistical analyses, and writing and editing of the manuscript.

17. Ma Y, Lin H, Wang Y, Huang H, **He X** (2021), A multi-stage hierarchical clustering algorithm based on centroid of tree and cut edge constraint, *Information Sciences* 557 (2021) 194–219.
18. Rutherford BR, Choi CJ, Choi J, Mass B., **He X.**, O'Boyle K., Sneed J.R., Brown P.J., Brickman A.M., Wall M.M., Roose S.P.(2020), Slowed processing speed disrupts expectancy-based placebo effects in late life depression. *The American Journal of Geriatric Psychiatry*, doi:<https://doi.org/10.1016/j.jagp.2020.11.001>
19. Merz E, **He X**, Myers B, Noble K. (2020). Socioeconomic Disadvantage, Chronic Stress, and Hippocampal Development in Children. *Neuroscience Insights* (15): 1-44, <https://doi.org/10.1177/2633105520931098>
20. Lichtin R, Merz E, **He X**, Desai P, Simon K, Melvin S, Maskus E, Noble K (2020), Material Hardship, Prefrontal Cortex-Amygdala Structure, and Internalizing Symptoms in Children, *Developmental Psychobiology* (63):364-377. <https://doi.org/10.1002/dev.22020>
21. Pagliaccio D, Cha J, **He X**, Cyr M, Yanes-Lukin P, Goldberg P, Fontaine M, Rynn M, Marsh R (2020). Structural neural markers of response to cognitive behavioral therapy in pediatric obsessive-compulsive disorder, *Journal of Child Psychology and Psychiatry* 61(12):1299-1308, PMID: 31889307
22. Morenob DR, Cykowicz Yael, Figner B, Wang Z, **He X**, Geronazzo-Alman L, Sun X, Cheslack-Postava K , Bisaga A, Hoven CW, Amsel L (2021). Delay discounting and neurocognitive correlates among inner city adolescents with and without family history of substance use disorder, *Developmental Cognitive Neuroscience*, 48:100942.
23. Merz, E.C., Desai, P.M., Rehman, R., Torres, S.D., Maskus, E.A., Melvin, S.A., Meyer, J., **He, X.**, & Noble, K.G. (2019). Socioeconomic disparities in chronic physiologic stress are associated with brain structure in children. *Biological Psychiatry*, 86(12):921-929.
24. Merz E.C., Maskus E.A., Melvin S., **He X.**, & Noble K.G. (2019). Socioeconomic disparities in language input are associated with children's language-related brain structure and reading skills. *Child Development*. 28 March 2019
25. Merz E.C., Maskus E.A., Melvin S., **He X.**, & Noble K.G. (2019). Parental punitive discipline and children's depressive symptoms: Associations with striatal volume. *Developmental Psychobiology*, 61(6):953-961.
26. Domakonda M, **He X**, Lee S, Cyr M, Marsh R (2019), Increased functional Connectivity between Ventral Attention and Default Mode Networks in Adolescents with Bulimia Nervosa, *Journal of the American Academy of Child and Adolescent Psychiatry*, 58(2):232-241.
27. Lv X, Ma Y, **He X**, Huang H, Yang J (2018), CciMST: A Clustering Algorithm Based on Minimum Spanning Tree and Cluster Centers, *Mathematical Problems in Engineering*, vol. 2018, Article ID 8451796.
28. Hou Z, Kong Y, **He X**, Yin Y, Zhang Y, Yuan Y (2018). Increased temporal variability of striatum region facilitating the early antidepressant response in patients

with major depressive disorder, *Progress in Neuropsychopharmacology & Biological Psychiatry* 85: 39-45

29. Merz EC, **He X**, Noble KG (2018), Anxiety, depression, impulsivity, and brain structure in children and adolescents, *NeuroImage: Clinical* 20: 243–251.
 30. **He X, Steinberg E, Stefan M, Fontaine, Simpson HB, Marsh R. (2018), Altered Frontal Inter-hemispheric and Fronto-limbic Structural Connectivity in Unmedicated Adults with Obsessive-Compulsive Disorder, *Human Brain Mapping*, DOI: 10.1002/hbm.23883.**
 This study used probabilistic tractography to investigate white matter structural connectivity in unmedicated adults with obsessive–compulsive disorder (OCD), free of comorbid conditions. Compared to matched healthy controls, individuals with OCD showed reduced fractional anisotropy (FA) in the forceps minor—which connects frontal interhemispheric regions—and in the right uncinate fasciculus (UNC), linking frontal and limbic areas. Importantly, lower FA in the forceps minor and decreased axial diffusivity in the UNC were both inversely correlated with OCD symptom severity, suggesting that disrupted frontal interhemispheric and fronto-limbic connectivity may underlie key features of OCD. These findings support a microstructural basis for altered frontal-limbic circuit function in OCD.
- I was a collaborator on the NIH R01 “Control and Reward Circuits as Targets for Repetitive Behaviors” (PI: Rachel Marsh) that produced the data used for this paper. I contributed to the DTI data preprocessing and analyses, statistical analyses, and writing and editing of the manuscript.
31. Yang J, Yin Y, Svob C, Long J, **He X**, Zhang Y, Xu Z, Li L, Liu J, Zhang Z, Wang Z, Yuan Y (2017), Atrophy of the Amygdala and its Functional Disconnection with the Cortico-striatal-pallidal-thalamic Circuit in Female major Depressive Disorder, *PLoS One* 12(1):e0168239.
 32. Liu Z, **He X**, Qiu Q, Liu F (2017), Class-Specific Random Forest With Cross-Correlation Constraints for Spectral–Spatial Hyperspectral Image Classification, *IEEE GEOSCIENCE AND REMOTE SENSING LETTERS*, Vol. 14, No. 2.
 33. Yin Y*, **He X***, Xu M, Hou Z, Song X, Sui Yu, Liu Z, Jiang W, Yue Y, Zhang Y, Liu Y, Yuan Y (2016), Structural and Functional Connectivity of Default Mode Network underlying the Cognitive Impairment in Late-onset Depression, *Scientific Reports*, DOI: 10.1038/srep376173 (**co-first author**)
 34. Piccolo LR, Merz EC, **He X**, Sowell ER, Noble KG (2016). Age-related changes in cortical thickness vary by socioeconomic status. *PLoS One* 11(9):e0162511.
 35. **He X**, Stefan M, Terranova K, Steinglass J, Marsh R (2016), Altered White Matter Microstructure in Adolescents and Adults with Bulimia Nervosa, *Neuropsychopharmacology*. 2016 (1-8). doi: 10.1038/npp.2015.354.
 36. Qiao J, Wang Z, Geronazzo-Alman L, Amsel L, Duarte C, Lee S, Musa G, Long J, **He X**, Doan T, Hirsch J, Hoven C (2015). Brain activity classifies adolescents with and without a familial history of substance use disorders. *Frontiers in Human Neuroscience*. 9:219. doi: 10.3389/fnhum.2015.00219.
 37. **He X**, Liu W, Li X, Li Q, Liu F, Rauh VA, Yin D, Bansal R, Duan Y, Kangarlou A, Peterson B, Xu D (2014), Automated Assessment of Diffusion Tensor Imaging Data Quality Using the Color Cast of Color-Encoded Fractional Anisotropy Images, *Magnetic Resonance Imaging*, 32(5): 446-456, PMID: 24637081.
 38. Lu Y, **He X**, Wen Y, Wang PS (2014), A new cow identification system based on iris analysis and recognition, *International Journal of Biometrics*, 6(1):18-32.

39. Li Q, **He X**, Wang Y, Liu H, Xu D, Guo F (2013), Review of spectral imaging technology in biomedical engineering: Achievements and challenges, *Journal of Biomedical Optics*, 18: 100901-1-100901-28. PMID: 24114019.
40. Li Q, Xu D, **He X**, Wang Y, Chen Z, Liu H, Xu Q, Guo F (2013), AOTF based molecular hyperspectral imaging system and its applications on nerve morphometry. *Applied Optics*, 52: 3891-3901. PMID: 23759836.
41. Hu S, Xu D, Peterson B, Wang Q, **He X**, Hu J, Xu X, Wei N, Long D, Huang M, Zhou W, Xu W, Zhang M, Xu Y (2013), Association of cerebral networks in resting state with sexual preference of homosexual men: A study of regional homogeneity and functional connectivity. *PLoS One* 2013, 8: e59426.
42. Li Q, Chen Z, **He X**, Wang Y, Liu H, Xu Q (2012), Automatic identification and quantitative morphometry of unstained spinal nerve using molecular hyperspectral imaging technology. *Neurochemistry International*, 61: 1375-1384. PMID: 23059447.
43. **He X**, Yang Z, Tsien J (2011), A hierarchical probabilistic model for rapid object categorization in natural scenes. *PLoS One*; 6: e20002.
44. Shen D, **He X**, Lu L (2009), A cattle iris segmentation method based on least square Principal. *Journal of Image and Graphics*. 14: 2132-2136. (Published in Chinese with English abstract).
45. An S, **He X**, Shi P (2008), Method for fake iris detection based on wavelet packet decomposition. *Measurement & Control Technology*; 27: 9-11. (Published in Chinese with English abstract)
46. **He X**, Yan J, Chen G, Shi P (2008), Contactless auto-feedback iris capture design, *IEEE Transactions on Instrumentation and Measurement*; 57: 1369-1375.
47. **He X**, Shi P (2007), A new segmentation approach for iris recognition based on hand-held capture device. *Pattern Recognition*; 40: 1326-1333.
48. **He X**, Shi P (2006), An efficient iris recognition system. *Journal of Computational Information Systems*; 2: 1269-1275.
49. **He X**, Shi P (2005), Enhancement method for low quality fingerprint images. *Journal of Data Acquisition & Processing*; 20: 440-443. (Published in Chinese with English abstract).
50. **He X**, Zhang P, Dai M, Huang J (2004), Digital watermarking technology based on Java. *Computer Applications and Software*; 21: 63-65. (Published in Chinese with English abstract).
51. **He X**, Huang J, Zhang P, Dai M, Zhang G (2004), A new fragile watermark with self-correcting. *Journal of Shanghai Normal University (Natural Sciences)* 2004; 33: 56-62. (Published in Chinese with English abstract).
52. **He X**, Huang J, Zhang G (2003), A fragile watermarking technology with self-correcting based on fractal compression. *Journal of Image and Graphics*; 8(Z1): 596-599. (Published in Chinese with English abstract)

Preprint / Under Review

53. Zheng X, Ravid O, Barry R.A.J, Kim Y, Wang Q, Kim Y, Zhu X, **He X** (2024), Denoising Variational Autoencoder as a Feature Reduction Pipeline for the diagnosis of Autism based on Resting-state fMRI, arXiv preprint arXiv:2410.00068. 2024 September. <https://doi.org/10.48550/arXiv.2410.00068>
54. Shatto LC, Thorp J, Friedman JS, Rosario I, Gu Z, Shi J, Zhuo M, Brown A, **He X***, Spagna A* (2023), Present: flexible neuroscience- and technology-driven frameworks for the study of classroom engagement, doi: <https://doi.org/10.1101/2023.06.23.544732> (***corresponding author**)

55. **He X**, Moreno D, Hou Z, Cheslack-Postava K, Jiang Y, Li T, Kishon R, Amsel L, Musa G, Wang Z, Hoven C (2023), Connectivity based Real-Time fMRI Neurofeedback Training in Youth with a History of Major Depressive Disorder, <https://arxiv.org/abs/2301.11846>
56. Tchibozo M, Kim D, Wang Z, **He X** (2022), Emotional Brain State Classification on fMRI Data Using Deep Residual and Convolutional Networks, <https://arxiv.org/abs/2210.17015>
57. Wengler K*, **He X***, Buck J, Taschereau-Dumouchel V, Kriegeskorte N, and Horga G (2024), Neural decoding and representational geometry of auditory verbal hallucinations (***co-first author**)
58. Friedman JS, Thorp J, Wang H, Wang R, **He X***, Spagna A* (2025), My Face Matters: Predicting Learning Engagement with Facial Expressions in an Undergraduate Seminar (***corresponding author**)

Invited and/or Peer-Selected Presentations, Conference papers

1. He X, Luo Y, Li R, Tong H, Thorp J, Spagna A, **He X**, A Deep-Learning Based Decoded EEG Neurofeedback Platform Using Muse-S, International Symposium on Decoded Neurofeedback, July 17-18, 2025, Nara, Japan. Poster.
2. **He X**, Gao Y, Wang Y, Wang X, Jiang Q, Xu B, Imaging Analysis of Calcium Activities in Brain Organoid Model of Neuropsychiatric Disorder, Brain Informatics 2024. December 13-15, Bangkok, Thailand. Short paper and oral presentation.
3. Wang Y, Wang Y, Meng Y, Xiao Y, **He X**, White Matter Integrity in Youth with Suicide Ideation/Suicide Attempt, Major Depressive Disorder, and Comorbidities: A Diffusion Tensor Imaging Study, Brain Informatics, 2024. Short paper and oral presentation.
4. Huang L Gong J, Xi Z, **He X**, Predicting fMRI Signals from Single-Channel EEG Using Deep Learning, Brain Informatics 2024. December 13-15, Bangkok, Thailand. Abstract and Oral presentation.
5. Tong H, Wang H, Friedman J, Thorp J, Spagna A, **He X**, A realtime EEG neurofeedback platform to predict students' attention level, Real-Time Functional Imaging and Neurofeedback meeting (rtFIN 2024), November 3-6, 2024, Heidelberg, Germany. Poster.
6. **He X**, Liu P, Wang Y, Zhang L, Liu Y, Lee S, Reduced White Matter Integrity in children with ADHD based on Diffusion Tensor Imaging (DTI) data from the ABCD Study, ABCD Insights & Innovations Meeting (AIIM). March 4-5, 2024, Bethesda, MD. Poster.
7. J. Friedman, SY Chuko, L. Penn, H. Shaik, CL. Lee Shatto, J. Thorp, X. Huang, H. Tong, **X. He**, A. Spagna, My Brain Matters: How Multimodal Intra-Individual Classifiers Reliably Predict Attention, While Inter-Individual Classifiers Do Not, April 13-26, Toronto, Canada, CNS 2024. Poster.
8. Zhou X, Shi J, He K, He S, Brown A, Spagna A, **He X**, Real time EEG brain classification using MUSE 2, Brain Informatics 2023, August 1-3, Stevens Institute of Technology, USA. Poster and Oral presentation
9. **He X**, Liu F, Lloyd C, Patel G, Marsh R, A pre-processing pipeline for DWI data acquired with diffusion gradient cycling on GE scanner, OHBM 2023, Montreal July 22-26, Canada. Poster.
10. M. A. Giebler, K. R. Simon, M. Amarente, E. C. Merz, X. He, K.G. Noble, Associations among SES, home language input, and resting-state functional connectivity in children, FLUX 2023, Santa Rosa, September 6-9, 2023. Poster.

11. Shatto C, Thorp J, Brown A, Friedman J, Shi J, Gu Z, **He X***, Spagna A*, Neuro-DBER: A flexible neuroscience- and technology-driven framework for the Scholarship of Teaching & Learning, Cognitive Neuroscience Society Annual Meeting 2023, San Francisco, March 25-28, 2023. Poster.
12. **He X**, A Decoded EEG Neurofeedback Platform Using Muse2, Real-Time Functional Imaging and Neurofeedback meeting (rtFIN 2022), Oct 17-21, 2022, New Haven, CT, USA. Oral presentation.
13. Shi J, He K, He S, Li J, Li M, Brown, Spagna A, **He X**, A Decoded EEG Neurofeedback Platform Using Muse2, Real-Time Functional Imaging and Neurofeedback meeting (rtFIN 2022), Oct 17-21, 2022, New Haven, CT, USA. Poster.
14. Zheng X, Ravid O, Kim Y, Zhu X, **He X**, A Denoising Variational Autoencoder for the diagnosis of Autism based on Resting-state fMRI, Data Science Day poster session, Columbia University, April 6, 2022, USA
15. Kim Y, Ravid O, Neria Y, **He X**, Zhu X, A Denoising Variational Autoencoder for identifying PTSD biotypes, Data Science Day, Columbia University, April 6, 2022, USA. Poster.
16. Spagna A, **He X**, Imaging the brains of students studying brain imaging: When Neuroscience Meets the class, Celebration of Teaching and Learning Symposium, Columbia University, March 29, 2022 USA. Oral presentation.
17. Shatto C, Gu Z, Hui X, Rosario I, Brown A, **He X**, Spagna A, Who gets an A+? Discipline Based Education Research meets Neuroscience in the Classroom, the 29th Annual Meeting of the Cognitive Neuroscience Society on April 23-26, 2022 in San Francisco, USA. Poster.
18. **He X**, Wengler K, Mihali A, Horga G, Decoding auditory verbal hallucinations based on patterns of neural activity in the temporal cortex in schizophrenia, the 60th Annual Meeting of the American College of Neuropsychopharmacology, December 5-8, 2021, Puerto Rico. Poster.
19. Tchibozo M, Wang Z, Kim D, **He X**, Emotional Brain State Classification on fMRI data using 3D Residual Neural Networks, DSI Health Analytics Center poster session, 2021.
20. Li M, Sun X, **He X**, EEGTransformer: Deep Learning based EEG to fMRI Transformation, Data Science Day 2021 Poster Session.
21. Merz E, **He X**, Simon K, Noble K, Parental Stress and Family Routines are Associated with Cerebral Blood Flow in Children, SOBP 2020. Abstract.
22. Simon K.R, Merz E.C, **He X**, Noble K.G Associations between Environmental Noise Exposure and Resting-State Functional Connectivity in Children. International Society for Developmental Psychobiology: Virtual Conference September 2020.
23. Simon K.R., Merz E.C., **He X.**, & Noble K.G. Environmental Noise, Language Skills, and Language-Related Brain Structure in Childhood. Flux Society: Virtual Conference. September 2020.
24. Wang W, Choi J, **He X** (2019), A Hierarchical Feature Extraction Pipeline using Resting-state fMRI for Autism Classification, The 2019 New York Scientific Data Summit (NYSDS), New York, NY. Oral presentation.
25. Feng Y, Chung E, Cycowicz Y, **He X** (2019), Motor Imagery Classification using Cascade ResNet-LSTM, The 25th Annual Meeting of the Organization for Human Brain Mapping (OHBM). Abstract.
26. **He X**, Moreno D, Li T, Tang H, Cycowicz Y, Wang Z, Amsel L, Musa J, Cheslack-Postava K, Bisaga A, Geronazzo-Alman L, Bresnahan M, Hoven (2019), Increased Functional Connectivity between DMN and FPN in Adolescents with Familial

- History of SUD, The 25th Annual Meeting of the Organization for Human Brain Mapping (OHBM). Abstract.
27. Choi J, **He X** (2019) , Classification of Autism based on Resting-state fMRI using Convolutional Neural Networks, The 25th Annual Meeting of the Organization for Human Brain Mapping (OHBM). Abstract.
 28. Choi J, **He X** (2018) , Classification of Autism Spectrum Disorder based on Brain Image using Convolutional Neural Networks, ITU/WHO workshop and Focus Group meeting on Artificial Intelligence for Health, Nov 14-16, New York, NY. Oral presentation.
 29. Domakonda M, **He X**, Lee S, Cyr M, Marsh R (2018), Increased Functional Connectivity Between Ventral Attention and Default Mode Networks in Adolescents with Bulimia Nervosa, AACAP's 65th Annual Meeting, October 22–27, Seattle, WA, Oral presentation.
 30. Lee YC, Chung EY, **He X** (2018), Brain State Classification on Functional Near Infrared Spectroscopy using Convolutional Neural Networks, Annual Data Science Town Hall Meeting, September 28–27, Columbia University, NYC, Poster.
 31. **He X**, Huang S, Liu Z, Moreno DR, Amsel L, Musa1 G, Hoven C (2018), Real-time Brain States Classification using 3D Convolutional Neural Networks, The 24th Annual Meeting of the Organization for Human Brain Mapping (OHBM), Jun 17-21, Singapore. Abstract.
 32. Merz EC, Maskus EA, **He X**, Meyer J, Noble KG (2018). The joint effects of socioeconomic disadvantage and parental depression on children's stress physiology and brain development. In M. Stallen and W. Molenmaker (Chairs), How neuroscience can contribute to solving societal problems. Symposium at the Association for Psychological Science (APS) Annual Convention, San Francisco, California, May. Poster.
 33. Desai PM, Merz EC, Maskus EA, **He X**, Noble KG (2018). Socioeconomic disadvantage, prefrontal cortical structure, and executive function in school-aged children. Poster presented at the Cognitive Neuroscience Society Annual Meeting, Boston, Massachusetts, March. Poster.
 34. **He X**, Huang SK, Moreno DR, Kishon R, Amsel L, Musa G, Noriega A, Wang Z, Hoven C (2017), Connectivity-based Real-time fMRI Neurofeedback in Youth with a History of Major Depression, *the Real-time Functional Imaging and Neurofeedback*, Nov. 29-Dec.1, Nara, Japan. Poster.
 35. Merz EC, Maskus EA, **He X**, Meyer J, Melvin S, Noble KG (2017). Family stress, hair cortisol, and children's amygdala and hippocampal volumes. *Talk given at the International Society for Developmental Psychobiology (ISDP) meeting*, Washington, D.C., November. Poster.
 36. Maskus EA, Merz EC, Desai P, **He X**, Noble KG (2017). Exploring links among subjective social status, prefrontal cortex structure, and cognitive skills in children. Poster presented at the Cognitive Development Society (CDS) Biennial Conference, Portland, Oregon, October. Poster.
 37. Noble KG, Piccolo L, Merz EC, Brito N, **He, X.** (2017). Socioeconomic inequality modifies links among cortical thickness, age, and cognition. In J. Booth (Chair), The effect of socioeconomic inequality on the brain depends on age, cognitive, and environmental factors. Paper symposium at the SRCD Biennial Meeting, Austin, Texas, April. Poster.
 38. Domakonda M, **He X**, Marsh , (2017), A Resting State Functional Magnetic Resonance Imaging (fMRI) Study of Attention Network Dysfunction in Adolescents

- With Bulimia Nervosa, *Journal of the American Academy of Child & Adolescent Psychiatry*, October 2017, Volume 56, Issue 10, Supplement, Page S160. Poster.
39. **He X**, Moreno DR, Liu Z, Amsel L, Musa G, Wang Z, Hoven C (2017), Amygdala response may distinguish participants with history of major depressive disorder, *The 23rd Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, June 25-29, Vancouver, Canada. Poster.
 40. **He X**, Stefan M, Fontaine M, Cha J, Simpson HB, Marsh R (2017), A DTI Quality control pipeline for running TRACULA, *The 23rd Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, June 25-29, Vancouver, Canada. Poster.
 41. Zhao W, Ma Y, **He X**, Huang H, Lin X, Zhang Y (2017), Face Liveness Detection Using Eulerian Video Magnification and SIFT Algorithm, *The 5th International Conference on Internet of Things and Information Technology (ICITIT)*, Nov 25-26, Chongqing, China.
 42. Lv X, Ma Y, **He X**, Huang H (2017), A Clustering Algorithm Based on MST and Cluster Centers, *the 2nd International Conference on Computer, Mechatronics and Electronic Engineering (CMEE)*, Dec 24-25, Xiamen, China.
 43. **He X**, Moreno DR, Geronazzo L, Amsel L, Hoven C, Wang Z (2016), Region of Interests in Local EPI Space for Real-time fMRI. *The 22nd Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, June 26-30, Geneva, Switzerland. Poster.
 44. **He X**, Geronazzo L, Amsel L, Moreno DR, Wang Z, Fan B, Musa G, Eisenberg R, Doan S, Wicks J, Bresnahan M, Hoven C (2016), White Matter Abnormalities in Individuals at High Risk of Depression. *The 22nd Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, June 26-30, Geneva, Switzerland. Poster.
 45. Rowny S, **He X**, Yang R, Zhou S, Prudic J, Javitt D, Hoven C (2016), Effects of ECT and MST on White Matter Integrity in Patients with Major Depression. *The 22nd Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, June 26-30, Geneva, Switzerland. Poster.
 46. **He X**, Stefan M, Fontaine M, Steinberg E, Simpson HB, Marsh R (2016). Tractography in Unmedicated Adults with Obsessive-Compulsive Disorder, *2016 Annual Meeting-Society of Biological Psychiatry (SOBP)*, May 12-14, Atlanta, Georgia, USA. Poster.
 47. **He X**, Stefan M, Fontaine M, Terranova K, Marsh R (2016), Using Tractography to Assess Structural Connectivity in Adolescents with Bulimia Nervosa, *2016 Annual Meeting-Society of Biological Psychiatry (SOBP)*, May 12-14, Atlanta, Georgia, USA. Poster.
 48. **He X**, Simpson HB, Marsh R (2015), White Matter Integrity in Unmedicated Adults with Obsessive-Compulsive Disorder, *2015 Annual Meeting-Society of Biological Psychiatry (SOBP)*, May 14-16, Toronto, Canada. Poster.
 49. **He X**, Stefan M, Terranova K, Steinglass J, Walsh T, Marsh R (2015), Analyses of White Matter Integrity in Adolescents and Adults with Bulimia Nervosa, *2015 Annual Meeting-Society of Biological Psychiatry (SOBP 2015)*, May 14-16, Toronto, Canada. Poster.
 50. **He X**, Geronazzo L.A, Lee S, Amsel L, Wang Z, Musa G.J, Duarte C.S, Doan T, Hirsh J, Hirsch, Hoven C (2015), White matter abnormalities in Adolescents with a Familial History of Substance Use Disorders. *The 21st Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, Jun.14-18, Hawaii, USA. Abstract.

51. **He X**, Liu F, Amsel L, Kangarlou A, Gerber A, Wang Z, Musa G, Geronazzo L, Duarte C, Duan Y, Doan T, Hirsch J, Hoven C (2015), A New Quality Control Pipeline for Diffusion Tensor Imaging. *The 21st Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, Jun.14-18, Hawaii, USA. Abstract.
52. **He X**, Liu F, Amsel L, Geronazzo L, Wang Z, Musa G, Duarte C, Kangarlou A, Gerber A, Duan Y, Qiao J, Doan T, Hirsch J, Hoven C (2014), Quality Criteria for DTI Data by Assessing Color-Encoded Fractional Anisotropy Images. *The 20th Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, Jun. 8-12, Hamburg, Germany. Poster.
53. **He X**, Yan X, Hao X, Bansal R, Wang Z, Peterson B, Xu D (2013), A New Volume Preserved Warping Method for Measuring Changes of Local Brain Volume, *the 19th Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, Jun.16-20, Seattle, WA, USA. Poster.
54. Parikh P, Wen Y, **He X**, Xu D, Peterson B, Delapaz L, Claassen J (2012), Feasibility of Advanced MR Imaging and Imaging Phenotype of Ictal-Interictal EEG Patterns in Critically Ill Patients with Nonstructural, Severe Brain Injury, *American Society of Neuroradiology (ASNR) 50th Annual Meeting & The Foundation of the ASNR Symposium*, New York, NY, USA. Poster.
55. Liu W, Liu X, **He X**, Zhou Z, Wen Y, Zhou Y, Peterson BS, Xu D: Spatial normalization of diffusion tensor images with voxel-wise reconstruction of the diffusion gradient direction, *2nd International Workshop on Multimodal Brain Image Analysis (MBIA) in conjunction with MICCAI* (pp. 134-146), 2012.
56. **He X**, Yang Z, Tsien J (2010), Object recognition based on hierarchically organized structures of natural objects, *Journal of Vision*, 10(7):989,2010, *the 10th Annual Meeting of Vision Sciences Society (VSS2010)*, *Journal of Vision*, 10(7):989, Naples, Florida, USA, May. Poster
57. **He X**, Yang Z, Tsien J (2010), Is categorizing objects in Natural scenes achievable? *Frontiers in Neuroscience 2010, Joint South East Nerve Net (SENN) and Georgia/South Carolina Neuroscience Consortium Conference (GASCNC)*, Atlanta, USA, Mar. Travel award. Oral presentation.
58. **He X**, Tsien J, Yang Z (2010), Object recognition based on hierarchically organized structures of natural objects. *Journal of Vision*; 10: 989.
<http://dx.doi.org/10.1167/10.7.789>. Abstract
59. **He X**, Lu Y, Shi P (2009), A new fake iris detection method, *Advances in Biometrics - 3rd International Conference on Biometrics*, (pp. 1139-1146), Alghero, Italy, June.
60. **He X**, Shi P (2008), Extraction of Complex Wavelet Features for Iris Recognition, *19th International Conference on Pattern Recognition (ICPR2008)*, Florida, USA, Dec.
61. **He X**, Lu Y, Shi P (2008), A fake iris detection method based on FFT and quality assessment, *Chinese Conference on Pattern Recognition*, (pp. 316-319), Beijing China, Oct.
62. Wan J, **He X**, Shi P (2007), An Iris Image Quality Assessment Method Based on Laplacian of Gaussian Operation, *IAPR Conference on Machine Vision Applications 2007 (MVA 2007)* (pp.248-251).
63. **He X**, An S, Shi P (2007), Statistical texture analysis-based approach for fake iris detection using support vector machines, *Advances in Biometrics - International Conference on Biometrics* (pp. 540-546).
64. **He X**, Shi P (2006), A novel iris segmentation method for hand-held capture device, *Advances in Biometrics - International Conference on Biometrics*, (pp. 479-485)

65. **He X**, Shi P (2005), An efficient iris segmentation method for recognition, *Pattern Recognition and Image Analysis - 3rd International Conference on Advances in Pattern Recognition*, (pp. 120-126)
66. **He X**, Shi P: An Eyelid Estimation Method for Iris Segmentation, *Regional Inter-University Postgraduate Electrical and Electronic Engineering Conference (RIUPEEEC 2005)*, Hong Kong, China, Jul. Oral presentation.